

Exploring Creepy Technology: Bridging Design, Context, and Experience

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Abstract

Technology now seamlessly integrates into our daily routines, from smart home assistants that control our lights to wearable fitness trackers monitoring our health. Reflecting on how technology usage has evolved from occasional convenience to daily necessity, it is key that interaction designers are intentional in not designing technology that is perceived as creepy. My PhD project examines how design qualities (e.g. modality, materiality) and use context shape perceptions of ‘creepy technology’, aiming to offer guidance to reduce unintended creepiness of technologies. Written for the CHI 2026 Student Mentoring Program, this paper outlines the purpose of my PhD project and the results of my research so far. This research aims to help designers create technology that users trust and welcome into their daily lives, with next steps including further investigating the nuanced boundaries of technological creepiness and completing my dissertation.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **HCI theory, concepts and models**.

Keywords

Interaction design, Creepy technology, Creepiness, User experience

ACM Reference Format:

Tora Jarsve. 2026. Exploring Creepy Technology: Bridging Design, Context, and Experience. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799200>

1 Introduction

Research on ‘creepy’ technology remains limited, particularly regarding everyday objects. McAndrew and Koehnke [11] highlighted this gap in psychology, questioning whether factors that make humans look creepy apply to objects as well. While creepiness is recognised in human experience, its underlying mechanisms have not been given sufficient attention, especially in design contexts

In HCI, creepy technology evokes unease, discomfort, or privacy invasion, appearing as wearables, installations, or systems with bodily interaction. These feelings arise from unpredictable

behaviour [20], fear or surveillance [14, 16], or both [19]. Examples include robots, AI, bionic limbs [18], and smart home sensors [3, 15].

While people may universally recognise something as creepy, the underlying reasons, intensity, and specific triggers often differ significantly between individuals due to personal factors. This variation in perception presents a particular challenge, as creepiness experiences are often difficult for both researchers and laypeople to articulate clearly. This complexity arises from the intricate entanglement of subjective experiences and objective conditions that constitute what we perceive as creepy. My research explores what design qualities (e.g. modality, materiality) of technologies are perceived as creepy, by whom and under what circumstances. My PhD investigates how appearance, behaviour, and data practices interact with contextual factors to shape creepiness perceptions, developing a conceptual framework and shared vocabulary to help designers and researchers articulate these mechanisms.

As the number of technologies surrounding us continues to increase, ensuring that these technologies are perceived positively becomes crucial, as first impressions significantly influence user acceptance [5, 12, 13]. Emerging technologies like GenAI have raised concerns among people regarding ethical considerations, accuracy, and appropriate regulatory frameworks for their development and use [4]. My work aims to support interaction designers navigate this nuanced landscape of subjective experience while still leveraging objective insights to design tomorrow’s technology responsibly.

2 Related Work

Creepy technology encompasses interactive systems that elicit the experience of creepiness in humans. In my PhD work, I build on different strands of research such as Woźniak et al.’s [19] conceptual model of creepiness and Yip et al.’s [20] research to understand creepiness as an umbrella term that contains different concepts such as *uncanny*, *eerie*, *uncomfortable*, *unpredictable*, and *undesirable*. While there is general implicit agreement among researchers about what creepy technology encompasses, we still face challenges in defining it without building on other related concepts. For instance, creepiness is often described using concepts like the uncanny valley and eeriness, which in turn use creepiness as a generative concept to design technologies intended to be uncanny and eerie. This circular relationship highlights the complexity of establishing a clear definition of creepiness independent of related concepts. My PhD project aims to address this by creating a conceptual framework.

One significant dimension of creepiness relates to appearance. Technologies that approach but do not fully achieve human likeness often fall into the uncanny valley [10], a concept particularly relevant for humanoid robots [2, 9, 10, 19]. When AI systems attempt



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CHI EA '26, Barcelona, Spain

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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799200>

to mimic human characteristics, they may also be described and explored through the lens of the uncanny valley.

While appearance is a significant factor in creepiness perception, other dimensions also contribute to the overall experience. Beyond the uncanny valley, technologies that exhibit unpredictability, create discomfort, or demonstrate undesirable characteristics are often perceived as creepy [19]. This extends to human augmentation technologies. The perception of human augmentation technologies offers valuable insight into the creepiness dimension of undesirable and unpredictable characteristics. Villa et al. [18] conducted a mixed-method study across four countries and identified the 'Peril' dimension as a key factor in how people perceive augmented humans. This research underscores how ethical considerations regarding the purpose and regulation of human augmentation technologies contribute to their perception as creepy, especially when these technologies blur the boundaries between human and machine in unpredictable ways.

On another note, the behaviour and movement of technologies can elicit feelings of creepiness. Technologies that exhibit unpredictability, create discomfort, or demonstrate undesirable characteristics are often perceived as creepy. This perspective extends beyond robots and AI-powered systems to include any type of technology. For example, the Soma Corset [8], a shape-changing garment that senses and responds to a user's breathing, creates an uncanny experience that is subjectively unpredictable — sometimes feeling like a supportive extension, sometimes like a monster, and other times like an organic other. It blurred the line between human and machine in deeply unsettling ways. The unpredictability of technology challenges users' ability to understand and trust the technology, contributing to its creepy nature, particularly when technologies create subjective experiences that defy easy categorisation as users struggle to reconcile the technology's programmed functions with their personal and often unsettling interpretations of its behaviour. Similarly, VR and AR technologies that behave in unexpected ways or create uncomfortable experiences can also fall under the umbrella of creepy technology, as demonstrated by Al-Zubeidi et al. [1] whom intentionally created a creepy VR environment to facilitate reflection regarding own experienced while walking along in a city at night.

Derived from Woźniak et al.'s [19] conceptual model of creepiness, is implied malice. This concept can be tied to data practices and hacking. Security and privacy concerns form a significant component of the creepy experience, where technologies that could potentially be used maliciously or that intrude upon personal privacy are perceived as creepy. This dimension connects to terms like undesirable and unpredictable, as technologies that might compromise security or privacy are often viewed with suspicion. Difference between the US and Europa with the General Data Protection Regulation (GDPR) regarding personal data privacy reflects societal sensitivity to technologies that might infringe on privacy, creating cultural differences in what is perceived as creepy. Contextual and subjective experiences and perceptions based on rules and regulations in the world such as this is usually not investigated in research on creepiness.

It is important to emphasise that not all technologies perceived as creepy are inherently harmful. Even though creepy technology might cause humans to feel uneasy, uncomfortable, or scared, such

technology does not always bypass privacy settings, collect excessive data, or cause actual harm. This distinction is crucial for developing a nuanced understanding of creepy technology that goes beyond simple categorisation as 'bad' or 'dangerous'. By exploring the broader ethical perspectives of creepy technology, we can better understand the complex relationship between technology design, user experience, and subjective perception.

2.1 Summary and Research Gaps

Research on creepy technology faces gaps in definitional clarity, as concepts like creepiness are often described using related terms like the uncanny valley and eeriness, creating circular definitions. Additionally, there is limited investigation into how contextual factors, cultural differences, and subjective experiences influence perceptions of creepiness, with insufficient focus on broader ethical perspectives beyond simple categorisation. My research addresses these gaps by developing a conceptual framework to establish clearer definitions, while conducting empirical studies examining how appearance, behaviour, and data practices interact with contextual factors across different technologies to understand when, by whom, and why technology is actually perceived as creepy.

3 Research Progress to Date

This section examines research challenges identified in my prior work and outlines my next steps focusing on both empirical investigations, as well as theoretical work to translate these insights for researchers and practitioners. The literature review addresses the definitional circularity in conceptualising creepiness, while the Furhat and ChatGPT study investigates how the appearance and behaviour of technologies contribute to the subjective experience of creepiness across different user groups.

3.1 Literature Review

I conducted a systematic literature review [7] (accepted to CHI26) to gain insight into the existing body of research on creepy technology. Our final corpus consisted of 115 papers. These papers were selected from an initial pool of 16,481 results, which were screened and reduced to 6,488 papers before applying the final inclusion criteria. The review focused on papers discussing the experience of creepiness, the design of and interaction with creepy technology, and concepts ranging from the uncanny valley effect to personalised advertisements. This literature review was conducted in line with established HCI literature review processes [17] and led to *The Creepiness Framework*.

The Creepiness Framework distinguishes between the *Structure of Creepiness*, which encompasses design-driven aspects under designers' control (such as contextual triggers, appearance elements, interactional dynamics, and narrative intent), and *Felt Creepiness*, which captures subjective user experiences shaped by personal histories, cultural contexts, and individual perceptions (see Figure 1). This framework is intended to help designers understand how and why technologies evoke feelings of unease, enabling them to create technologies that are less likely to trigger unintended creepy experiences while providing insights for critical and reflective design practices.

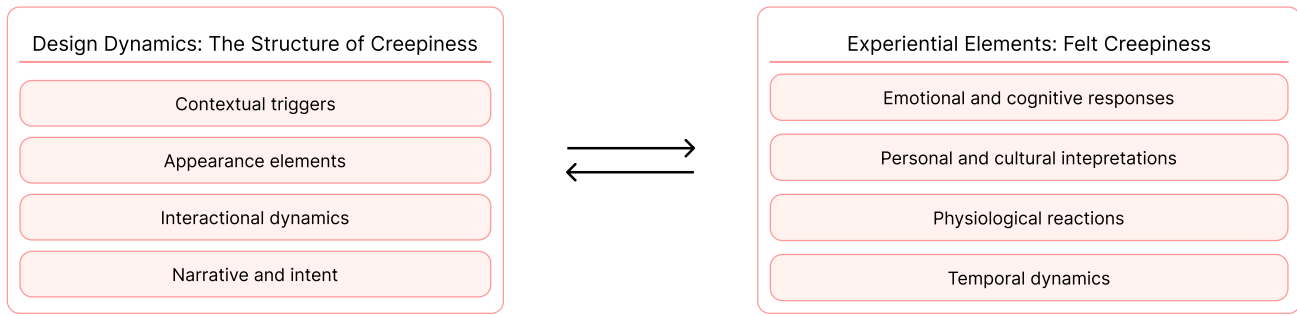


Figure 1: The Creepiness Framework, mapping the interplay between design dynamics (*Structure of Creepiness*) and experiential elements (*Felt Creepiness*) [7]. The *Structure of Creepiness* captures design-driven aspects such as contextual triggers, appearance elements, interactional dynamics, and narrative and intent while *Felt Creepiness* reflects the emotional and cognitive, personal and cultural, physiological, and temporal responses of users. Together, these categories demonstrate that creepiness emerges not from design features or user interpretations in isolation, but from their dynamic interplay.

3.1.1 Research Challenges. Based on the systematic review, we identified several aspects that impact our understanding of creepy technology in HCI. These aspects include definitional clarity, methodological approaches, prototype development, and temporal understanding.

Definitional Challenges. Analysis of 115 papers reveals substantial inconsistencies in how ‘creepiness’ is conceptualised, with 75.6% providing some form of definitions while 24.3% used the concept without clarification. However, within the group of papers providing a definition, no clear consensus could be identified. This definitional inconsistency challenges establishing unified theoretical foundations and cross-study comparisons, hindering the development of cohesive design principles to address and mitigate feelings of creepiness.

Methodological Limitations. The research landscape faces methodological challenges, particularly with quantitative approaches dominating. This creates difficulties capturing the nuanced, subjective experience of creepiness, which involves emotional, cognitive, cultural, and physiological responses inadequately measured quantitatively. HCI research on creepiness has relied heavily on questionnaires, and experimental psychology paradigms that often overlook the rich, situated experiences defining this phenomenon.

Prototype Development Limitations. Analysis of prototype proximity to the body reveals a significant research gap in exploring embodied technology interactions. This creates challenges in understanding how body-integrated technologies evoke creepiness. While existing tools like the PCTS [19] and SHAPE scale [18] have advanced creepiness measurement, they focus on specific use cases, leaving a broader gap in understanding how technologies generate unease.

Absence of Longitudinal Research. The most significant methodological challenge is the absence of longitudinal research, hindering understanding of how creepiness perceptions evolve over time. Feelings may intensify or diminish with repeated interactions due to

the novelty effect, where initial encounters with unfamiliar technologies amplify discomfort that later diminishes with familiarity. Without longitudinal studies, we cannot distinguish between transient discomfort and enduring perceptions or contextual influences over time.

3.2 Exploring User Expectations and Perceived Creepiness in AI: A Study on Furhat and ChatGPT

As AI technology continues to expand its capabilities and become more integrated into daily life, user expectations are rising, yet mismatches between these expectations and actual AI behaviour can lead to discomfort, unease, and perceptions of creepiness. A key factor influencing these perceptions is the physical appearance of AI interfaces, which can trigger the uncanny valley effect or create conflicting emotional responses. To address this research gap, particularly regarding how appearance affects perceived creepiness, I conducted a user study on Furhat and ChatGPT [6].

Our mixed-method study ($n = 20$) involved participants randomly assigned to interact with either ChatGPT or Furhat, using video observations, interviews, conversation logs, and The Perceived Creepiness of Technology Scale (PCTS) [19] to gather comprehensive data through triangulation. Three main themes emerged from the thematic analysis:

- **Human-Like Attributes:** Both AI systems were well-received for their human-like qualities. Furhat’s speech understanding and active listening created positive surprise and interest, while ChatGPT’s human-like interactions (e.g. its ability to follow a natural conversational flow, transitioning smoothly between topics) provided comfort, facilitating natural conversations.
- **Machine-Like Attributes:** Machine traits generated negative reactions. Both systems were perceived as intrusive when asking personal questions, creating tension. Furhat’s physical form made personal discussions uncomfortable, while

ChatGPT's written interface felt unnatural for personal interaction.

- **Interface Impact:** Conversation dynamics affected comfort differently. Furhat participants disliked interruptions, while ChatGPT's fast pace pressured responses. Paradoxically, AI was criticised for both lacking sufficient human-like qualities and being too human-like. Interface perceptions varied: Furhat was entertaining yet disconcerting, while ChatGPT proved more frustrating.

Quantitative findings showed both systems elicited creepiness perceptions, with Furhat perceived as significantly more creepy, particularly in unpredictability and undesirability.

3.2.1 Research Challenges. The study reveals key challenges for developing acceptable AI systems:

Design Interface Challenges. Physical robots and text-based interfaces require different design principles due to their distinct characteristics. Systems must respect user boundaries regarding personal information and incorporate expectation alignment mechanisms. Design approaches should intentionally balance human-like and machine-like attributes rather than maximising human-likeness.

Behavioural Pattern Challenges. AI behaviours like quick responses create user pressure, necessitating turn-taking mechanisms that avoid exploiting cognitive vulnerabilities. Research is needed on sustainable long-term human-AI relationships, as users naturally apply human social norms to AI interactions.

Subjective Experience Challenges. Systems should proactively shape appropriate user expectations based on limited information. Current measurement tools focus primarily on negative experiences, requiring more comprehensive assessment methods. Context-aware design approaches are needed to accommodate diverse user preferences and situational factors.

3.3 Future Research Focus

As my overarching research aim is to understand which design qualities of technologies are perceived as creepy, by whom, and under what circumstances these perceptions emerge, I conduct work addressing different aspects of this phenomenon. Addressing several of the research challenges identified through my previous work — like the behavioural pattern challenge, design interface challenge and the lack of focus on the subjective experience — my current research explores the multifaceted nature of user experiences with different technologies.

3.3.1 Research Strand 1. Currently, I am examining how personalised advertising mechanisms and voice-activated robotic systems might trigger uncomfortable and creepy responses, seeking to understand the underlying factors that transform ordinary technological interactions into experiences we label as unsettling.

3.3.2 Research Strand 2. My ongoing interest centres on the temporal dimension of user experiences, exploring how perceptions evolve over time and how we might develop strategies to address emerging discomfort in technologies that were initially perceived

as unproblematic. This actively addresses the absence of longitudinal research I identified through the literature review. Through all my empirical investigations, I aim to map the complex terrain that constitutes this phenomenon while encompassing everything from interface design choices to ethical considerations, from immediate emotional responses to long-term societal implications, addressing the definitional challenge also identified in the literature review.

3.3.3 Research Strand 3. Looking forward, I hope to develop immersive virtual reality (VR) prototypes that deliberately evoke complex emotional responses as a means to stimulate critical reflection on significant societal challenges, positioning technological discomfort not merely as a negative byproduct but as a potential catalyst for deeper engagement with critical topics. This will actively address the definition challenge, as I will need to find the right approach to creepy to have the intended affect. In addition, I will also be addressing the methodological limitations, as I will be most interested in the rich descriptions of the subjective experience in my quest to pinpointing the right *type* of creepiness.

Acknowledgments

I would like to thank my supervisors, Jasmin Niess and Suhas Govid Joshi for their continuous guidance and support. I would also like to thank Nadine Wagener for additional guidance, support and feedback.

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